

## Part A

Table 1.

1. Anomalies in database are inconsistencies or errors that can occur when data is inserted, deleted or updated, It can occur due to mistakes or error in data collection. Anomalies are unexpected and abnormal.

2. **INSERTION ANOMALY:** It occurs when extra information is to be added. For example in above table to insert a course the student should be known, if a student is not known a new course can't be inserted and as primary key can't be NULL.

**UPDATE ANOMALY:** If we change stdclass of student or the stdcity then we need to update all the tuples of student

**DELETION ANOMALY:** If ~~a new~~ It occurs when we delete a record and fact is deleted like here if we delete 3rd row we will lose information about courseNo c3 because it was only one entry against this tuple.

Table 2:

1. **INSERTION ANOMALIES:** If a new branch is to be inserted with no e staff initially it can't be inserted because staffNo can't be null as it is primary key. and each time we enter a staff we have to enter branch details again and an error can result in data inconsistency, redundancy.

**UPDATE ANOMALY:** let say we want to update branch address or telNo then we have to update all the tuple with that branch. error can lead to wrong data.

**DELETION ANOMALY:** if we delete staff from branch with one staff let say branch B004 or B003 so upon deletion of the staff the branch data will also be lost.

2.

1NF

STAFF BRANCH

| <u>staffNo</u> | name | position | salary | branchNo | branchAddress | telNo |
|----------------|------|----------|--------|----------|---------------|-------|
|----------------|------|----------|--------|----------|---------------|-------|

2NF

STAFF

| <u>StaffNo</u> | name | position | salary | branchNo |
|----------------|------|----------|--------|----------|
|----------------|------|----------|--------|----------|

BRANCH

| <u>branchNo</u> | branchAddress | telNo |
|-----------------|---------------|-------|
|-----------------|---------------|-------|

3NF

It is already in 3NF as no transitive dependency exist.

## Part B

### Question 1:

1NF

PROJECT

|                     |               |                 |                |
|---------------------|---------------|-----------------|----------------|
| <u>Project code</u> | project title | project Manager | Project Budget |
|---------------------|---------------|-----------------|----------------|

EMPLOYEE

|              |                    |               |               |                 |             |
|--------------|--------------------|---------------|---------------|-----------------|-------------|
| project code | <u>employee No</u> | employee Name | department no | department name | hourly Rate |
|--------------|--------------------|---------------|---------------|-----------------|-------------|

2NF

PROJECT

|                     |               |                  |                |
|---------------------|---------------|------------------|----------------|
| <u>Project code</u> | project title | project managers | project budget |
|---------------------|---------------|------------------|----------------|

EMPLOYEE

|                    |               |               |                 |
|--------------------|---------------|---------------|-----------------|
| <u>employee No</u> | employee name | department No | department name |
|--------------------|---------------|---------------|-----------------|

EMPLOYEE PROJECT

|                     |                    |             |
|---------------------|--------------------|-------------|
| <u>project code</u> | <u>employee No</u> | hourly Rate |
|---------------------|--------------------|-------------|

3NF

PROJECT

|                     |               |                 |                |
|---------------------|---------------|-----------------|----------------|
| <u>Project Code</u> | project title | project Manager | project Budget |
|---------------------|---------------|-----------------|----------------|

EMPLOYEE

|                    |               |               |
|--------------------|---------------|---------------|
| <u>employee No</u> | employee Name | department No |
|--------------------|---------------|---------------|

EMPLOYEE PROJECT

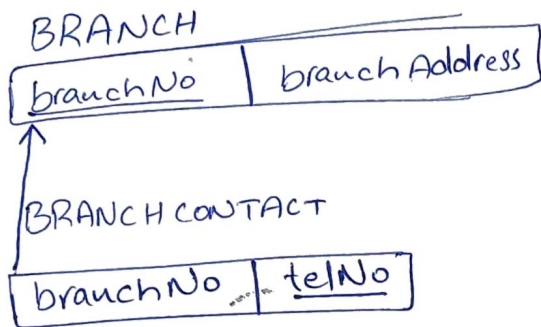
|                     |                    |             |
|---------------------|--------------------|-------------|
| <u>Project Code</u> | <u>employee No</u> | hourly Rate |
|---------------------|--------------------|-------------|

DEPARTMENT

|                      |                 |
|----------------------|-----------------|
| <u>department No</u> | department Name |
|----------------------|-----------------|

## Question no: 2

1. The table is not in 1NF because telNos is multivalued attribute.
2. 1NF, 2NF, 3NF are same



3. Primary key  $\Rightarrow$  branchNo & telNo  
Foreign Key  $\Rightarrow$  branchNo from BRANCH CONTACT TABLE  
Alternative Key  $\Rightarrow$  branchAddress.

## Question 3

1. The table is in 1NF
2. It is not in 2NF, branch Address is ~~partial~~ not Fully dependent, we have to remove partial dependency now.

3. 1NF STAFF ALLOCATION

|         |          |               |      |          |              |
|---------|----------|---------------|------|----------|--------------|
| staffNo | branchNo | branchAddress | name | position | hoursPerWeek |
|---------|----------|---------------|------|----------|--------------|

2NF STAFF

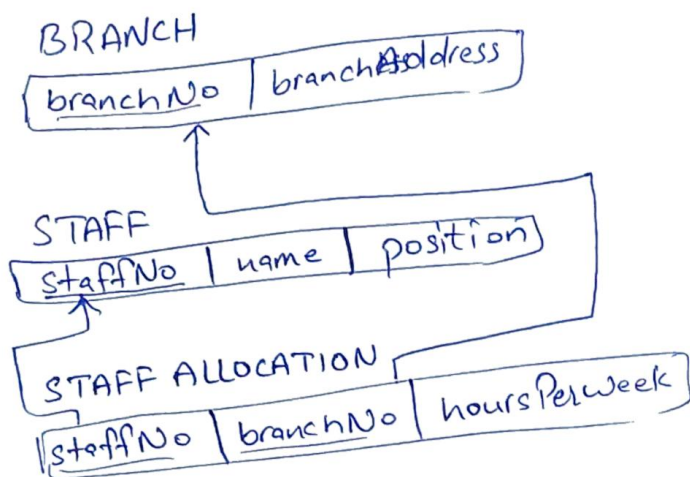
|         |      |          |              |          |
|---------|------|----------|--------------|----------|
| staffNo | name | position | hoursPerWeek | branchNo |
|---------|------|----------|--------------|----------|

An arrow points from the **branchNo** attribute in the 2NF STAFF table to the **branchNo** attribute in the BRANCH table below.

BRANCH

|          |               |
|----------|---------------|
| branchNo | branchAddress |
|----------|---------------|

## 3NF



4. Primary keys  $\Rightarrow$  BRANCH(branchNo), STAFF(staffNo)  
Foreign Key  $\Rightarrow$  STAFF ALLOCATION(staffNo, branchNo)  
Composite Key  $\Rightarrow$  STAFF ALLOCATION(staffNo, branchNo)

## Question 4:

- The table is not in 3NF because it has transitive dependency.  
As branchAddress, telNo, mgrStaffNo, name can be accessed with branchNo  
branchNo and name can also be accessed with mgrStaffNo.  
 $\text{branchNo} \rightarrow \text{mgrStaffNo}$   
 $\text{mgrStaffNo} \rightarrow \text{name}$
- 

~~3NF~~ already in 1NF

~~3NF~~ Assuming branchNo & mgrStaffNo as composite Key

3NF

| BRANCH        |
|---------------|
| branchNo      |
| branchAddress |
| telNo         |
| mgrStaffNo    |

MANAGER STAFF

|            |
|------------|
| mgrStaffNo |
| name       |

3. Primary Key  $\Rightarrow$  branchNo in Branch Table & mgrStaffNo in Manager Staff Table  
Foreign Key  $\Rightarrow$  mgrStaffNo in Branch Table

## Questions 5

1NF:

EMPLOYEE

|              |          |             |
|--------------|----------|-------------|
| <u>Empid</u> | emp-name | emp-address |
|--------------|----------|-------------|

~~emp~~

CONTACT ~~DETAIL~~

|        |            |
|--------|------------|
| emp-id | emp-mobile |
|--------|------------|

2NF: There is no partial dependency, so it is in 2NF

3NF: There is no transitive dependency, so it is in 3NF

## Question 6

i) ~~Modifying~~ the event coordinator name ~~name~~ in events table, it requires to change multiple tuples which can lead to data inconsistency if not done properly.

SOLUTION: use organizer id instead of event coordinator name.

ii) It can lead to problems when using ~~Join~~ joins. else no anomalies because it has no relation, but in join it can have some ~~unexpected~~ unexpected data.

iii) Deleting ~~and~~ an organizer in event organizer can result in deletion of event.  
It can be solved by splitting the table

iv) ~~Let say if new event has been proposed but the event coordinator~~  
In event organizer table adding a new organizer managing multiple events will duplicate record.

- v) All table are in 1<sup>st</sup> Normal Form if Participants table is not NULL & it needs to have participant id.
- vi) ~~Event Organisers is not in 2NF~~

~~Organisers ID~~  
~~Society Table~~

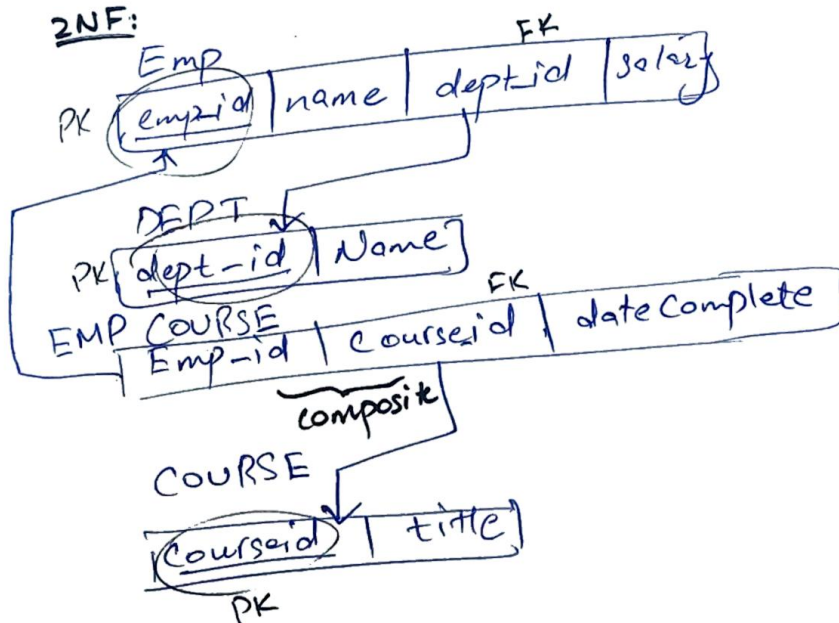
vi) All are in 2NF

vii) Society Table is not in 3NF it has non key attribute The

## Question 7

1. Yes it is in 1NF
2. because it has no multivalued attribute
3. No
4. Partial dependence exists

2NF:

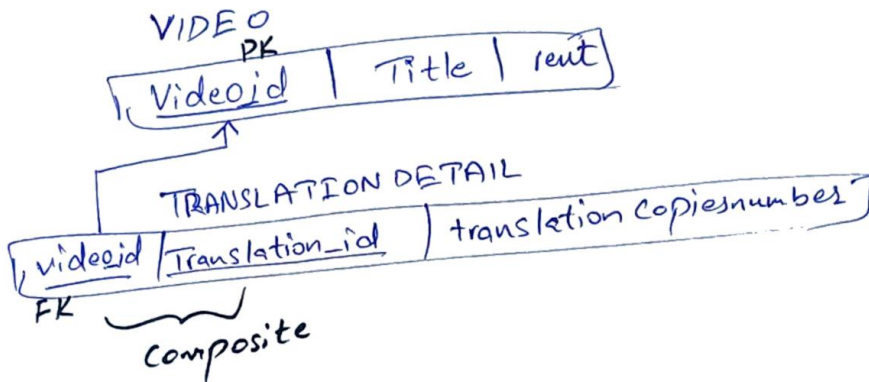


5. Marked on above schema. Labelled above.
6. Yes it is in 3NF
7. Because no transitive dependency exists.
8. Same Primary Keys as in 2NF.

### Question 8:

1. No
2. Assuming Translation\_id and video\_id as composite key then it is not in 2NF.

2NF:

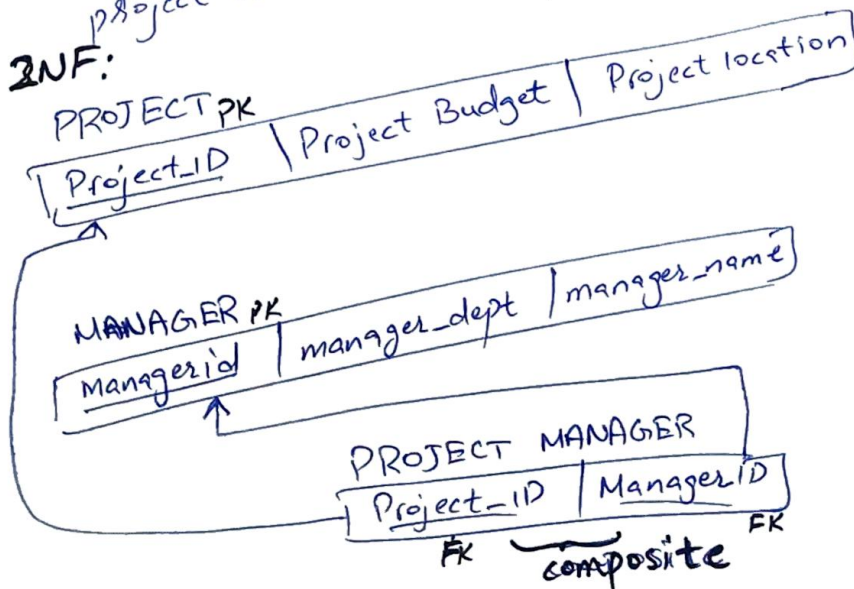


3. Labelled above
4. Yes it is in 3NF
5. because no transitive dependencies.
6. some Key's as in 2NF, labelled above

### Question 9:

1. No the table is not in 2NF
2. because partial dependence is there, because manager name depend on Manager ID and can also be accessed by project ID.

2NF:



3. Key's labelled in schema representation
4. Yes they are in 3NF
5. because no transitive dependency occurs
6. same as in 2NF.